

General Directions

1. Use legible/clear handwriting
2. For your answers use the separate answer sheet provided.
3. Use uppercase letters for your answers (for Parts I through V).
4. Never use pencils to write your answers. Corrections are forbidden for any work done with a pencil.
5. Make sure that this exam consists of EIGHT printed pages.

Part I. Write 'TRUE' if the statement is correct or 'FALSE' if it is incorrect and write down your answer on the separate answer sheet provided.

1. Sweat glands are examples of exocrine glands.
2. Poikilotherms bask in the sun when they get too hot.
3. People who look healthy can infect you with HIV/AIDS.
4. Deforestation may lead to CO₂ build up in the atmosphere.
5. The deepening of voice is a female's secondary sexual characteristic.

Part II. Match the premises under the column "A" with the responses given under the column "B" by writing the letter of your choice on the separate answer sheet provided.

Column 'A'

6. It prevents implantation and pregnancy
7. It contains the female hormone oestrogen
8. It is the surgical removal of a fallopian tube.
9. It is a thin rubber cap inserted into the vagina.
10. It prevents sperm from getting into the semen

Column 'B'

- A. The mini-pill
- B. The mixed pill
- C. The vasectomy
- D. The diaphragm
- E. The female condom
- F. The intrauterine device
- G. The female sterilization

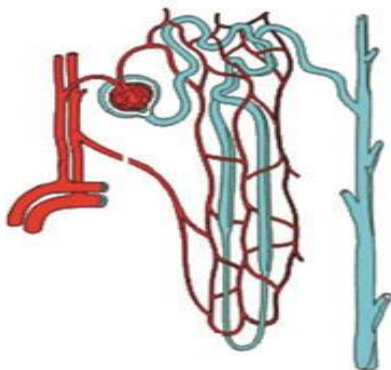
Part III. Fill in the blank spaces with the most appropriate words on the separate answer sheet provided.

11. The removal of an amine group from a molecule is called _____.
12. A hormone that lowers the blood glucose level is said to be _____.
13. The type of summer sleep, performed by cold-blooded animals is called _____.
14. The waterproof upper surface layer found in many types of leaf is known as _____.
15. The hormone which stimulates egg release in women and testosterone production in men is called _____.

Part IV. Choose the correct answer from the alternatives given and write the letter of your choice on the separate answer sheet provided.

16. What is homeostasis? It is the process in which the
 - A. internal conditions are kept variable.
 - B. internal temperature is kept changeable.
 - C. body maintains normal function and stability.
 - D. body fails to adjust itself to the environment.
17. Which of the following is NOT a physiological method of temperature regulation in homoiotherms?
 - A. Panting and licking
 - B. Shivering or sweating
 - C. Seeking shelter or shade
 - D. Constricting or dilating vessels
18. The temperature deep inside our body is referred as ...
 - A. variable temperature.
 - B. extreme temperature.
 - C. ambient temperature.
 - D. core body temperature.
19. The leaf is flat and wide giving...
 - A. large surface area to collect light and long distances for gases to diffuse.
 - B. small surface area to collect light and long distances for gases to diffuse.
 - C. large surface area to collect light and short distances for gases to diffuse.
 - D. small surface area to collect light and short distances for gases to diffuse.
20. In one of the steps to test for starch in a leaf, a student dipped a leaf in hot water. Why? To
 - A. soften the leaf.
 - B. remove chlorophyll.
 - C. make the leaf brittle.
 - D. detach the leaf from its stalk.

21. The following diagram shows a nephron in a human kidney.



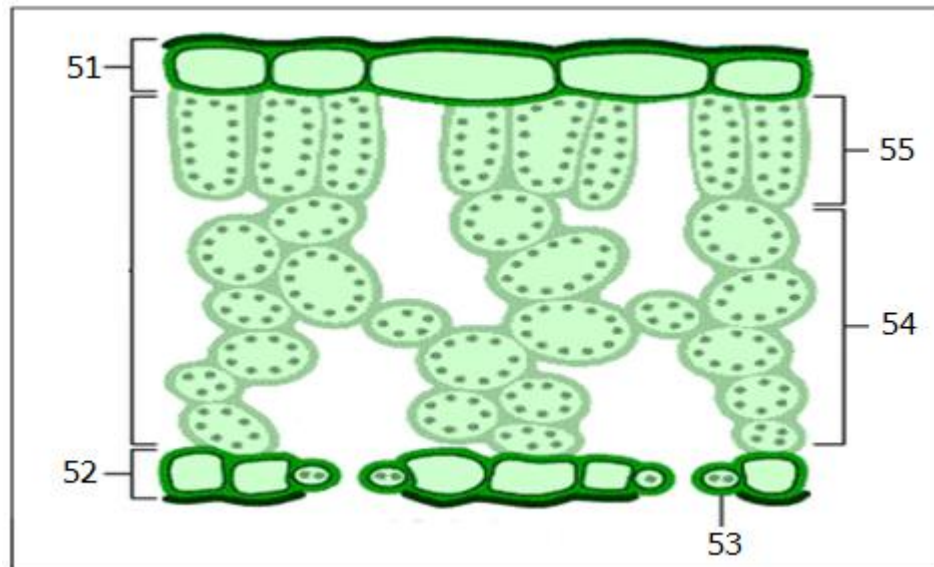
- What is the name of a product that is produced by the above structure? The
- A. bile
B. urine
C. enzyme
D. hormone
22. What is the tuft or cluster of capillaries that brings blood to the kidney for filtration? The
- A. glomerulus.
B. loop of Henle'.
C. proximal tubule.
D. glomerular capsule.
23. What should a biologist do in order to remove chlorophyll from the leaf? She
- A. boils the leaf in water.
B. boils the leaf in alcohol.
C. washes the leaf with water.
D. washes the leaf with glycerine.
24. Which one indicates the chemical equation for photosynthesis?
- A. $6\text{CO}_2 + 6\text{H}_2\text{O} \xrightarrow{\text{using sunlight}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$
B. $6\text{CO}_2 + \text{C}_6\text{H}_{12}\text{O}_6 \xrightarrow{\text{using sunlight}} 6\text{H}_2\text{O} + 6\text{O}_2$
C. $6\text{CO}_2 + 6\text{O}_2 \xrightarrow{\text{using sunlight}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O}$
D. $6\text{O}_2 + 6\text{H}_2\text{O} \xrightarrow{\text{using sunlight}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{CO}_2$
25. Which one of the following hormones secreted by the thyroid glands regulates the rate of oxidation in cells?
- A. Glucagon
B. Thyroxine
C. Adrenaline
D. Progesterone
26. Which of these statements is INCORRECT regarding the function of hormones?
- A. For development and growth
B. To maintain body temperature
C. To maintain of internal environment
D. For reproduction and sexual differentiation

27. Endocrine glands differ from exocrine glands in that
- A. endocrine glands release hormones, whereas exocrine glands release waste.
 - B. endocrine glands are all interconnected; whereas exocrine glands act completely independently.
 - C. endocrine glands are formed by epithelial tissue, but exocrine glands are primarily connective tissue.
 - D. endocrine glands are ductless and exocrine glands release secretions at the body's surface or into ducts.
28. The secretions from which of the following glands differ between males and females?
The
- A. adrenal.
 - B. gonadal.
 - C. pancreas.
 - D. parathyroid.
29. Which hormone prepares the body for stress situations?
- A. Thyroxin
 - B. Glucagon
 - C. Adrenaline
 - D. Testosterone
30. What causes the onset the puberty in males?
- A. An increase in the secretion of testosterone by testis.
 - B. A decrease in the secretion of testosterone by testis.
 - C. An increase in the secretion of progesterone by ovary.
 - D. A decrease in the secretion of progesterone by ovary.
31. Use the items (i-v) below to answer the next question.
- i. Still air
 - ii. Dry air
 - iii. Humid air
 - iv. Hot temperature
 - v. Windier condition
- Which of the following will increase the rate of transpiration?
- A. i, ii, and iii
 - B. i, iii and v
 - C. iii, iv and v
 - D. ii, iv and v
32. According to Ethiopian Institute of Biodiversity Conservation (IBC), which mammal species is at the brink of extinction?
- A. Walia ibex
 - B. Gelada baboon
 - C. Menelik's Bushbuck
 - D. Swayne's Hartebeest
33. Which National Park in Ethiopia houses Mountain Nyala?
- A. Omo National Park
 - B. Nechisar National Park
 - C. Bale Mountain National Park
 - D. Yangudi Rassa National Park

34. Of the following endemic species in Ethiopia, which one is used for the extraction of vegetable oil?
- A. *Eragrostis teff*
 - B. *Ficus vasta* Forssk
 - C. *Ensete ventricosum*
 - D. *Guizotia abyssinica*
35. Which Ethiopian Emperor is said to have organized the collection of seeds of many indigenous trees from Wof-Washa forest? Emperor
- A. Menelik II
 - B. Zera Yakob
 - C. Yohannes IV
 - D. Hale Selassie I
36. Which of the following refers to non-renewable resources?
- A. Trees
 - B. Cattle
 - C. Fossil fuels
 - D. Crop plants
37. Fruits which are unripened when they are picked and shipped to market can quickly be ripened by exposure to
- A. auxin.
 - B. ethylene.
 - C. cytokinin.
 - D. gibberillin.
38. Auxins cause a plant to bend towards light because they
- A. concentrate in the dark side of cells to elongate those cells.
 - B. are lost through photosynthesis as more light strikes the plant.
 - C. are lost by means of transpiration as more light strikes the plant.
 - D. increase in amount on the lit side of the plant to lengthen the cells.
39. Unlike tropisms, nastic movements are in response to
- A. wind.
 - B. darkness.
 - C. directional stimuli.
 - D. non-directional stimuli.
40. Plants take up mineral ions in the soil by means of
- A. osmosis.
 - B. simple diffusion.
 - C. active transport.
 - D. facilitated diffusion.
41. Air pollution with chlorofluorocarbons (CFC's) have caused
- A. acid rain.
 - B. ozone hole.
 - C. global cooling.
 - D. volcanic eruption.
42. The strong attraction force between water and xylem tissue is
- A. adhesion.
 - B. cohesion.
 - C. capillarity.
 - D. root pressure.
43. The rate of flow of water through the xylem is regulated by
- A. the turgidity of guard cells.
 - B. the force of transpiration pull.
 - C. the availability of water in soil.
 - D. passive transport by the phloem.

44. Which of the following is the most important factor for seed germination?
- A. Sunlight
 - B. Carbon dioxide
 - C. Adequate water
 - D. Extreme temperature
45. One of the following is true about germination of monocot seeds?
- A. Leaves emerge from cotyledon
 - B. Cotyledon remains below the ground
 - C. Cotyledon appears above the ground
 - D. Radicle pushes cotyledon above the ground
46. Which of the following play an opposite role in recycling carbon between living and the environment?
- A. Respiration and combustion
 - B. Evaporation and condensation
 - C. Photosynthesis and respiration
 - D. Condensation and photosynthesis
47. Which of the following structure is responsible for the absorption of water from the soil?
- A. Xylem
 - B. Cortex
 - C. Phloem
 - D. Root hair
48. What two structures helps an elephant to survive in hot, dry climates by increasing the surface area through which heat can be lost? The
- A. wide domed head
 - B. long black, shaggy hair
 - C. huge curved tusks and trunk
 - D. wrinkles in the skin and the big, thin ears
49. Select a plant tissue that is attacked by insect pests such as aphids by sticking their mouth parts.
- A. Xylem
 - B. Phloem
 - C. Parenchyma
 - D. Collenchyma
50. Which one of the following factors increases the rate of transpiration?
- A. Dry air
 - B. Still air
 - C. Humid air
 - D. Lower temperature

Part V. The diagram below illustrates the cross section of a leaf. Identify the parts designated by the numbers 51-55 and write down your answer on the separate answer sheet provided.



Part VI. Read the following questions carefully and write down brief answers on the separate answer sheet provided.

56. What are goose-bumps?
57. Describe the word hypothermia?
58. How would you manage if you have a mild form of diabetes?
59. What is the importance of the light-dependent and light-independent reactions in photosynthesis?
60. What are the two aspects which substantiate that kidneys are vitally important in maintaining homeostasis? Explain.